

Critical Summary: Decentralized Detection with Many Sensors

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1 Introduction - Decentralized Remote Hypothesis Testing

In our critical summary, we discuss the results of the paper “Decentralized Detection with Many Sensors: Optimality of Exchangeable and Identical Encoding Policies” [1]. The paper addresses the problem of Decentralized Hypothesis Testing: many sensors observe an environment, compress their observations into actions, and send those actions to a fusion center (FC), which decides between two hypotheses, H_1 and H_2 . While the paper’s authors build tools for both the finite and infinite number of sensors regimes, in our work we focus on the finite- N setting.

2 Problem Setup

Suppose there are two hypotheses, H_1 and H_2 ; the true one is modeled as H^* , a random variable with $P(H^* = H_1) = p$, $P(H^* = H_2) = 1 - p$. The system setup is:

- $N < \infty$ sensors, each observing the environment and producing $y^i \in \mathcal{Y}^i$, the observation of the i -th sensor, for $i \in \{1, \dots, N\}$.
- Each sensor quantizes its observation using a mapping $\gamma^i : \mathcal{Y}^i \rightarrow \mathcal{U}^i$, where $\mathcal{U}^i$ denotes the set of actions the sensor can send. γ^i is the decision rule / policy of the i -th sensor.
- The FC’s policy is noted as $\gamma^0 : \mathcal{U}^0 \times \mathcal{U}^1 \times \dots \times \mathcal{U}^N \rightarrow \{1, 2\}$. Essentially, the fusion center decides $u^0 = \gamma^0(u^{1:N})$, $u^0 \in \{1, 2\}$. Denote the set Γ^i as the set of policies of sensor i , Γ^0 as the set of policies for the FC.
- The cost function we will focus on is:

$$c(H_j, u^0) = \begin{cases} 1 & \text{if } u^0 \neq j \\ 0 & \text{if } u^0 = j \end{cases} \quad (1)$$

- Our goal is to minimize the expected cost:

$$J^N(\gamma^{0:N}) \triangleq \mathbb{E}^{\gamma^{0:N}}[c(H, u^{0:N})] \quad (2)$$

2.1 Assumptions

In order to make the problem tractable, the authors of the paper consider a few assumptions:

- (i) $y^1, \dots, y^N$ are i.i.d. conditioned on H
- (ii) Under either hypothesis, sensor i 's observation has a probability density, measured against one common reference measure shared across both hypotheses.
- (iii) $\mathcal{U}^i = \mathcal{U}$ for $i = 1, \dots, N$, and $\mathcal{U}$ is a finite set.
- (iv) The likelihood ratio:

$$l^i = L(y^i) := \frac{f_{y^i|H}(y^i|H_2)}{f_{y^i|H}(y^i|H_1)} \text{ a.s.} \quad (3)$$

is well-behaved and invertible.

- (v) Define $I_N^{KCR} \triangleq \{u^{1:N}, \gamma^{1:N}\}$ to be the Known Common Randomization set, which includes all actions and policies of the sensors. We assume this set is known to the FC.

While (i) is a common assumption in order to simplify the math and make it tractable, assumptions (ii) and (iv) are essential in order for l^i to be a sufficient statistic of y^i , as will be discussed later. Assumption (iii) dictates that all sensors have the same action set, an underlying assumption about the compression and rate being used. This will be discussed further in section *, where we will offer some modifications for the solution setup. Assumption (v)'s essentiality will be discussed in Section 3.3.

3 Sufficient Statistics, Existence of an Optimal Encoding Policy

3.1 Exponential Decay of the Error Probability

According to the law of large numbers, we can expect that $J^N(\gamma^{0:N})$ will converge to the probability of error P_e under decision rules $\gamma^{0:N}$. Also, under the criterion of minimizing the probability of error, using the MAP rule at the fusion center is optimal for any sensor policy [2].

[Proposition 5.1, 2] - If $y^1, \dots, y^N$ are i.i.d. given H^* and $P_{Y|H_1} \neq P_{Y|H_2}$, then there exists a decision rule γ' and constants $\alpha, \beta > 0$ independent of N such that if all sensors use γ' and the FC uses MAP then: $P_e < \alpha e^{-N\beta}$.

From [Proposition 5.1, 2], we can conclude that:

$$J^{N*}(\gamma^{0:N}) = \inf_{\gamma^{0:N}} \{J^N(\gamma^{0:N})\} \leq J^N(\gamma^{0:N}) < \alpha e^{-N\beta} \quad (4)$$

From Equation (4) we can understand that P_e of any relevant sequence of policies is exponentially decaying as N grows large, hence we need to study and optimize the **Error Exponent**, which characterizes how fast the error will decay given a sequence of policies. This leads to the definition:

$$J_{EE}^N(\gamma^{0:N}) \triangleq -\frac{1}{N} \log(J^N(\gamma^{0:N})) \quad (5)$$

We should notice that $J^N(\gamma^{0:N}) \leq 1$ is a probability measure meaning its log is negative. By adding the negative sign, $J_{EE}^N(\gamma^{0:N})$ becomes a strictly positive rate. Hence, our goal is to maximize $J_{EE}^N(\gamma^{0:N})$ to achieve the fastest possible exponential decay of the error probability.

3.2 l^i is a Sufficient Statistic

l^i is a sufficient statistic of H : once its value is known, nothing else about the raw observation y^i can further move one's belief about which hypothesis is true. With only two hypotheses under consideration, the entire way in which the observation's density can vary with H is exhausted by a single comparison – how large the density is under H_2 relative to how large it is under H_1 , at that particular observation. A single number is therefore inherently enough to hold everything that changes with H , and that number is, by construction, exactly l^i . (How this shows up concretely inside the paper's own machinery – via Lemma 1 – is worth treating separately in a later section.)

No information loss: a Markov-chain and DPI reading

Sufficiency means that any two raw observations producing the same l^i are, for the purpose of distinguishing H_1 from H_2 , interchangeable: whatever detail of y^i is discarded in forming l^i carries no evidential weight about H . This has a clean information-theoretic statement. Since l^i is a deterministic function of y^i , the chain

$$H \rightarrow y^i \rightarrow l^i$$

is trivially Markov, and the Data Processing Inequality (DPI) gives

$$I(H; l^i) \leq I(H; y^i). \quad (6)$$

Equality holds *iff* l^i is sufficient for H – equivalently, *iff* $H \rightarrow l^i \rightarrow y^i$ is *also* a Markov chain, i.e. y^i carries no further information about H once l^i is fixed. This gives equality:

$$I(H; l^i) = I(H; y^i). \quad (7)$$

3.3 Optimality of Threshold Policies and the MAP Decoder

Lemma 1

For any fusion center decision rule γ^0 ,

$$\inf_{\gamma^{1:N} \in \Gamma^N} J^N(\gamma^0, \gamma^{1:N}) = \inf_{\gamma^{1:N} \in \Gamma_{TS}^N} J^N(\gamma^0, \gamma^{1:N}). \quad (8)$$

where Γ_{TS}^N is the class of threshold policies.

Why this reduction holds. One direction is immediate: $\Gamma_{TS}^N \subseteq \Gamma^N$, so restricting the infimum to threshold policies can only raise it,

$$\inf_{\gamma^{1:N} \in \Gamma^N} J^N(\gamma^0, \gamma^{1:N}) \leq \inf_{\gamma^{1:N} \in \Gamma_{TS}^N} J^N(\gamma^0, \gamma^{1:N}). \quad (9)$$

For the reverse direction, fix sensors $j \neq i$ to *arbitrary* policies and ask what sensor i 's best response is. As shown below, this best response is always a threshold rule – regardless of what the other $N - 1$ sensors are doing, since the sub-problem sensor i faces turns out to be linear in its own likelihood ratio l^i . Because this holds for any fixed configuration of the rest, we can replace each sensor's policy, one at a time, with its threshold best-response: the cost J^N never increases at any step. After N such replacements every sensor uses a threshold rule, and the resulting cost is no larger than that of the original arbitrary profile. Taking the infimum over the starting profile gives

$$\inf_{\gamma^{1:N} \in \Gamma_{TS}^N} J^N(\gamma^0, \gamma^{1:N}) \leq \inf_{\gamma^{1:N} \in \Gamma^N} J^N(\gamma^0, \gamma^{1:N}), \quad (10)$$

and combined with the trivial direction, equality holds. This is the standard *person-by-person optimization* argument: joint optimality follows from unilateral optimality of each coordinate, holding all others fixed.

Concretely, fixing every sensor's policy except sensor i 's, sensor i 's optimization is:

$$\inf_{u^i \in \mathcal{U}} \left\{ \sum_{j=1,2} P(H_j | y^i) E_{\gamma^{-i*}} [c(H_j, \gamma^0(u^{1:N})) | H_j] \right\} \quad (11)$$

By Bayes' rule,

$$P(H_j | y^i) = \frac{f_{y^i|H}(y^i | H_j) P(H_j)}{f_{y^i}(y^i)}, \quad j = 1, 2, \quad (12)$$

and taking the ratio of the two, the normalizer $f_{y^i}(y^i)$ cancels:

$$\frac{P(H_2 | y^i)}{P(H_1 | y^i)} = \frac{f_{y^i|H}(y^i | H_2)}{f_{y^i|H}(y^i | H_1)} \cdot \frac{P(H_2)}{P(H_1)} = l^i \cdot \frac{P(H_2)}{P(H_1)}. \quad (13)$$

So the ratio of posteriors – all that matters for comparing the two terms in the sum above – depends on y^i only through the likelihood ratio l^i : exactly the sufficiency of l^i from Section 3.2, showing up again inside the optimization itself.

Writing $g^i(H_j, u^i) \triangleq E_{\gamma^{-i*}} [c(H_j, \gamma^0(u^{1:N})) | H_j]$ for the expected cost of action u^i under H_j given the other sensors' fixed policies, and factoring the positive term $P(H_1 | y^i)$ out of the sum (it does not affect the arg inf),

$$\sum_{j=1,2} P(H_j | y^i) g^i(H_j, u^i) = P(H_1 | y^i) \left[g^i(H_1, u^i) + \frac{P(H_2 | y^i)}{P(H_1 | y^i)} g^i(H_2, u^i) \right]. \quad (14)$$

Dropping the positive prefactor $P(H_1 | y^i)$ and substituting the ratio derived above, the subproblem reduces to

$$\inf_{u^i \in \mathcal{U}} \left[g^i(H_1, u^i) + g^i(H_2, u^i) l^i \frac{P(H_2)}{P(H_1)} \right]. \quad (15)$$

This is linear in l^i for each action, so the best action is whichever line is lowest at that value of l^i , switching only at the points where lines cross – a threshold rule.

Lemma 2

Under the assumptions above, for any deterministic encoding policy of the sensors, the optimal policy of the fusion center is given by:

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) \geq t \\ H_2 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) < t \end{cases} \quad (16)$$

for some $t \in \mathbb{R}$ and $\Delta_N(u^{1:N}; \gamma^{1:N})$ given by:

$$\Delta_N(u^{1:N}; \gamma^{1:N}) \triangleq \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i | H_1)}{P^{\gamma^i}(u^i | H_2)} \quad (17)$$

Recalling that in [2] it was established that the MAP rule is optimal for any deterministic sensor policy. The MAP rule for our problem is:

$$\gamma^{0*}(u^{1:N}) = \arg \max_{i \in \{1,2\}} P(H_i|u^{1:N}) \quad (18)$$

It can be written in the form of a log likelihood decision rule:

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} \geq 0 \\ H_2 & \text{if } \log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} < 0 \end{cases} \quad (19)$$

And looking at the log likelihood:

$$\log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} = \log \frac{P(u^{1:N}|H_1)}{P(u^{1:N}|H_2)} + \log \frac{P(H_1)}{P(H_2)} \quad (20)$$

$$= \log \prod_{i=1}^N \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} + \underbrace{\log \frac{p}{1-p}}_{\triangleq -Nt} \quad (21)$$

$$= \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} - Nt \quad (22)$$

Here, Equation (20) follows from Bayes' rule, and Equation (21) from y^i being i.i.d. given H . We should just note that:

$$\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} - Nt \geq 0 \iff \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \geq t \quad (23)$$

And this sums up how the optimal decision rule of the FC is established. It is important to note that this decision rule is why assumption (v) is important for the paper's solution: in order to calculate $\Delta_N(u^{1:N}; \gamma^{1:N})$, the FC must know how u^i was derived, to give it its proper weight in the decision process.

3.4 Existence of an Optimal Solution

Lemmas 1 and 2 characterize the necessary structure of an optimal policy – threshold at the sensors, MAP at the FC. Theorem 1, using the compactness result of Lemma 3, establishes that a minimizer with this structure exists.

Lemma 3

Γ_{TS}^N is compact under total variation.

Lemma 3 is an important step towards establishing the **existence** of an optimal solution, the paper's main contribution to the finite N problem.

Theorem 1

There exists an optimal decision rule sequence $\gamma^{0:N*} = (\gamma^{0*}, \gamma^{1*}, \dots, \gamma^{N*})$. These policies are MAP for the FC and threshold for the sensors.

The proof of Theorem 1 follows from Lemmas 1 and 2, together with the compactness of Γ_{TS}^N (Lemma 3). Since $J^N(\gamma^{0:N})$ is continuous in the sensors' policies over this compact set, Weierstrass's extreme value theorem guarantees that the infimum is attained.

4 Calculating the error exponent

Under assumptions (i) - (v) and Theorem 1, we can formulate a closed-form bound for the error exponent. Bounding each of the two error events with a Chernoff bound, in terms of a policy-dependent Chernoff coefficient $M(\alpha, \gamma^i)$, and combining them gives a bound on P_e . Taking $-\frac{1}{N} \log(\cdot)$ of this bound and letting $N \rightarrow \infty$ yields, for K distinct policies used by fractions c_k of the sensors:

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0,1]} \sum_{k=1}^K c_k \log M(\alpha, \gamma^k) \quad (24)$$

Where

$$M(\alpha, \gamma^i) \triangleq \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i | H_2)^{1-\alpha} P^{\gamma^i}(u^i | H_1)^\alpha \quad (25)$$

The full derivation is given in Appendix 6.1.

We can use the bound in Equation (24) to get a closed form bound of the exponent in Equation (4). Assuming all the sensors policies are identical, $K = 1, c_1 = 1$ and we get the definition of **Chernoff Distance** [3]:

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0,1]} \log M(\alpha, \gamma) = C(P_{Y|H_1}^\gamma, P_{Y|H_2}^\gamma)$$

Discussion of the result

$$C(P_{Y|H_1}^\gamma, P_{Y|H_2}^\gamma) = - \min_{\alpha \in [0,1]} \log \left(\sum_{u \in \mathcal{U}} P^\gamma(u^i | H_2)^{1-\alpha} P^\gamma(u^i | H_1)^\alpha \right) \quad (26)$$

As defined in [3], the Chernoff distance (Chernoff information) is a measure of **distinguishability** between two probability distributions;

If $P(u^i | H_1) = 1$, the hypotheses are perfectly distinguishable, so $P(u^i | H_2) = 0$: the sum inside the logarithm vanishes, the error exponent diverges, and $P_e \rightarrow 0$.

On the other hand, if $P(u^i | H_1) = P(u^i | H_2)$ for all $u \in \mathcal{U}$, the hypotheses are indistinguishable: the sum equals 1, the error exponent vanishes, and P_e fails to decay.

For any other conditional probabilities, the Chernoff distance provides a solid bound on the error exponent.

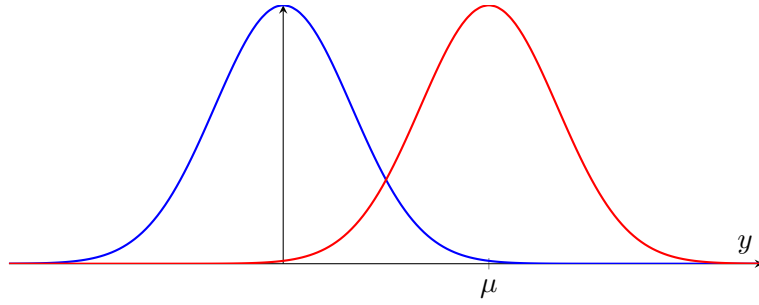
This result is highly intuitive: as the quantized observations under each hypothesis become more distinguishable, the Chernoff distance grows. This yields a larger error exponent, which drives a steeper exponential decay in the probability of error. In practical terms, greater distinguishability at the sensor level guarantees an exponentially better detection performance as the number of sensors grows.

5 Considering rate

We now turn to the total rate consumed by the sensors. We consider two encoding and transmission policies. The first is the **vanilla** policy, under which every sensor transmits its action using R bits regardless of how confident it is in that decision. The second is a **silent** policy, in which we introduce two thresholds on each sensor's decision rule: a sensor transmits its action using R bits only if its observation falls outside of a silent bin, meaning it is "confident" in its decision.

Assumptions

- Assumptions (i) to (v) hold.
- Gaussian model - $y^i|H_1 \sim \mathcal{N}(0, \sigma^2)$ and $y^i|H_2 \sim \mathcal{N}(\mu, \sigma^2)$



- Each sensor can send up to R bits to the fusion center, meaning $\mathcal{U} = \{u^1, \dots, u^{2^R}\}$.
- All sensors use the same policy - γ^V for the Vanilla setup and γ^S for the silent setup.

For each setup, we derive the corresponding Chernoff distance bound of Equation (26) and evaluate the setups' performance via simulation.

General R -bit Threshold Policy

We generalize Setup A and Setup B into a single family of threshold policies, indexed by the number of bits R , a bin width $\Delta > 0$, and a silence half-width $\delta \geq 0$.

Let $L \triangleq 2^R$ and $\mathcal{U} = \{u^1, \dots, u^L, u^S\}$, where u^S denotes silence. For $\delta \geq 0$, define the thresholds

$$t_j(\delta) \triangleq \begin{cases} \frac{\mu}{2} - \delta - \left(\frac{L}{2} - j\right)\Delta, & j = 1, \dots, \frac{L}{2} \\ \frac{\mu}{2} + \delta + \left(j - \frac{L}{2} - 1\right)\Delta, & j = \frac{L}{2} + 1, \dots, L \end{cases}, \quad t_0 \triangleq -\infty, \quad t_{L+1} \triangleq +\infty. \quad (27)$$

The policy $\gamma_{\Delta, \delta}^R$ sends u^m if $y \in (t_{m-1}, t_m)$ for $m = 1, \dots, L/2$, stays silent if $y \in (t_{L/2}, t_{L/2+1})$, and sends u^m if $y \in (t_m, t_{m+1})$ for $m = L/2 + 1, \dots, L$.

At $\delta = 0$, $t_{L/2} = t_{L/2+1} = \mu/2$: the silence interval degenerates to a point, u^S is never sent, and $\gamma_{\Delta, 0}^R$ is an L -region always-transmit policy. We define

$$\gamma^V \triangleq \gamma_{\Delta, 0}^R, \quad \gamma^S \triangleq \gamma_{\Delta, \delta}^R, \quad \delta > 0. \quad (28)$$

Standardizing by σ ,

$$a_j \triangleq \frac{t_j(\delta)}{\sigma}, \quad c_j \triangleq \frac{t_j(\delta) - \mu}{\sigma} = a_j - s, \quad s \triangleq \frac{\mu}{\sigma}, \quad a_0 = c_0 \triangleq -\infty, \quad a_{L+1} = c_{L+1} \triangleq +\infty. \quad (29)$$

Every quantity below is written in terms of a_j, c_j and applies to γ^V and γ^S alike.

Symbol Probabilities

Since $L(y)$ is increasing in y (Appendix 6.2), each region's probability under the Gaussian model is a difference of Q -functions. For the left-hand symbols $m = 1, \dots, L/2$,

$$P^{\gamma_{\Delta, \delta}^R}(u^m | H_1) = Q(a_{m-1}) - Q(a_m), \quad P^{\gamma_{\Delta, \delta}^R}(u^m | H_2) = Q(c_{m-1}) - Q(c_m), \quad (30)$$

for the silence symbol,

$$P^{\gamma_{\Delta, \delta}^R}(u^S | H_1) = Q(a_{L/2}) - Q(a_{L/2+1}), \quad P^{\gamma_{\Delta, \delta}^R}(u^S | H_2) = Q(c_{L/2}) - Q(c_{L/2+1}), \quad (31)$$

and for the right-hand symbols $m = L/2 + 1, \dots, L$,

$$P^{\gamma_{\Delta, \delta}^R}(u^m | H_1) = Q(a_m) - Q(a_{m+1}), \quad P^{\gamma_{\Delta, \delta}^R}(u^m | H_2) = Q(c_m) - Q(c_{m+1}). \quad (32)$$

Threshold Symmetry

The thresholds (27) satisfy, for every $j = 0, \dots, L + 1$,

$$t_{L+1-j}(\delta) = \mu - t_j(\delta), \quad \text{equivalently} \quad a_{L+1-j} = s - a_j, \quad c_{L+1-j} = -a_j. \quad (33)$$

Together with $f_{y|H_1}(y) = f_{y|H_2}(\mu - y)$, Eq. (33) gives

$$P^{\gamma_{\Delta, \delta}^R}(u^m | H_1) = P^{\gamma_{\Delta, \delta}^R}(u^{L+1-m} | H_2), \quad m = 1, \dots, L, \quad (34)$$

and the silence probability is hypothesis-independent,

$$P^{\gamma_{\Delta, \delta}^R}(u^S | H_1) = P^{\gamma_{\Delta, \delta}^R}(u^S | H_2) \triangleq P^{\gamma_{\Delta, \delta}^R}(u^S) = Q\left(\frac{s}{2} - \frac{\delta}{\sigma}\right) - Q\left(\frac{s}{2} + \frac{\delta}{\sigma}\right). \quad (35)$$

Chernoff Coefficient

Generalizing Equation (25) to $L + 1$ symbols,

$$M(\alpha; R, \Delta, \delta) \triangleq P^{\gamma_{\Delta, \delta}^R}(u^S) + \sum_{m=1}^L P^{\gamma_{\Delta, \delta}^R}(u^m | H_2)^{1-\alpha} P^{\gamma_{\Delta, \delta}^R}(u^m | H_1)^\alpha. \quad (36)$$

By Equation (24) with $K = 1$,

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0,1]} \log M(\alpha; R, \Delta, \delta). \quad (37)$$

Substituting $m \rightarrow L + 1 - m$ and using Eqs. (34)-(35) gives $M(1 - \alpha; R, \Delta, \delta) = M(\alpha; R, \Delta, \delta)$. Each summand in (36) is convex in α , so $M(\cdot; R, \Delta, \delta)$ is convex on $[0, 1]$; symmetry about $\alpha = 0.5$ then places its minimum there, as in Equations (27) and (29).

Closed-Form Bound

$$C(R, \Delta, \delta) \triangleq -\log M(0.5; R, \Delta, \delta)$$

$$= -\log \left[P^{\gamma_{\Delta, \delta}^R}(u^S) + \sum_{m=1}^L \sqrt{P^{\gamma_{\Delta, \delta}^R}(u^m | H_1) P^{\gamma_{\Delta, \delta}^R}(u^m | H_2)} \right], \quad (38)$$

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq C(R, \Delta, \delta). \quad (39)$$

Recovering Setup A and Setup B

Define $C_V(R, \Delta) \triangleq C(R, \Delta, 0)$ and, for $\delta > 0$, $C_S(R, \Delta, \delta) \triangleq C(R, \Delta, \delta)$.

At $\delta = 0$, $a_{L/2} = a_{L/2+1} = s/2$, so $P^{\gamma^V}(u^S) = 0$ and (38) reduces to the L -region vanilla bound. At $R = 1$ ($L = 2$), $a_1 = s/2 \mp \delta/\sigma$ does not depend on Δ , and (38) reduces to Equation (28) at $\delta = 0$ and to Equation (30) at $\delta > 0$, recovering the one-bit case exactly.

The Unquantized Ceiling

Before evaluating the bound it is worth fixing what it is bounded by. Suppose the sensor forwards y^i itself, at infinite precision. Under the Gaussian model

$$M_\infty(\alpha) \triangleq \int f_{y|H_2}^{1-\alpha}(y) f_{y|H_1}^\alpha(y) dy = \exp\left(-\alpha(1-\alpha)\frac{\mu^2}{2\sigma^2}\right), \quad (40)$$

which is again symmetric about $\alpha = 1/2$ and convex, so the minimum sits there as in Equation (38), giving

$$C_\infty \triangleq -\min_{\alpha \in [0,1]} \log M_\infty(\alpha) = \frac{\mu^2}{8\sigma^2} = \frac{s^2}{8}. \quad (41)$$

Since $H \rightarrow y^i \rightarrow u^i$ is Markov for any policy, the Data Processing Inequality of Section 3.2 applies to γ as it did to l^i : quantization can discard evidence about H , never create it. Hence

$$C(R, \Delta, \delta) \leq C_\infty \quad \text{for every } R, \Delta, \delta, \quad (42)$$

and C_∞ is the natural yardstick against which every finite-rate policy is measured. Note that C_∞ depends on the model only through $s = \mu/\sigma$: the whole problem is how many standard deviations apart the two hypotheses sit.

Transmitted Rate

The parameter R is the rate γ^V pays, but not the rate γ^S pays: a silent sensor sends nothing. Using Equation (35), the expected transmitted rate is

$$\bar{R}(R, \Delta, \delta) \triangleq R \left(1 - P^{\gamma_{\Delta, \delta}^R}(u^S)\right), \quad (43)$$

so $\bar{R} = R$ for γ^V and $\bar{R} < R$ for γ^S . Comparisons made at equal R therefore charge γ^S for bandwidth it never uses.

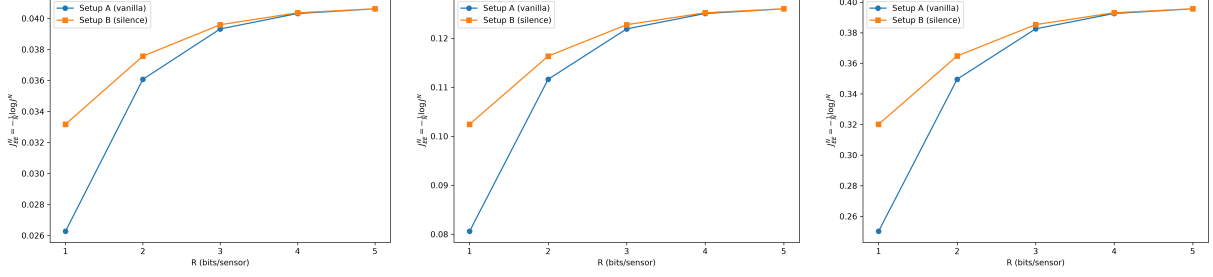


Figure 1: Normalized error exponent J_{EE}^N against R at the optimal (Δ^*, δ^*) , for $N = 3000$ and $\text{SNR} = -5, 0, +5$ dB (left to right). Setup A is γ^V , Setup B is γ^S .

Simulation Results

We evaluate $\gamma_{\Delta, \delta}^R$ for $R = 1, \dots, 5$ at $N = 3000$ sensors and $p = \frac{1}{2}$, at three operating points $\text{SNR} \in \{-5, 0, +5\}$ dB. At each point (Δ^*, δ^*) are chosen to maximize the closed-form bound $C(R, \Delta, \delta)$ of Equation (38); Setup A pins $\delta = 0$, so γ^V optimizes over Δ alone. The fusion center computes J^N exactly, from which J_{EE}^N follows by Equation (5).

Figure 1 shows one shape repeated at all three operating points. J_{EE}^N rises monotonically in R and saturates onto the unquantized ceiling C_∞ of Equation (41), as Equation (42) requires. Saturation is fast: a single bit already recovers roughly 63% of C_∞ under γ^V and 81% under γ^S , and by $R = 3$ both policies sit within a few percent of the ceiling. The entire design question therefore lives at $R \leq 3$; past that the extra bits buy nothing.

The silence policy dominates the vanilla one at every R , as $C_S > C_V$ predicts, but the whole of that advantage is concentrated at low rate. It is largest at $R = 1$ and by $R = 4$ the two curves are indistinguishable. The optimized geometry explains why. Table 1 lists (Δ^*, δ^*) against R : the cell width Δ^* decays geometrically, by a factor of roughly 0.55 per added bit, while the ratio δ^*/Δ^* stays close to 0.4 throughout. The silence bin holds an essentially fixed fraction of a cell, so it shrinks with the cell; by Equation (35) the silence probability vanishes with it, and γ^S degenerates continuously into γ^V . The two effects visible at the knee of Figure 1 – silence ceasing to help, and quantization ceasing to cost – are the same phenomenon, $\Delta^* \rightarrow 0$, seen from two sides.

R	2	3	4	5	6	7
Δ^*	0.988	0.567	0.321	0.179	0.099	0.054
δ^*	0.384	0.224	0.127	0.070	0.038	0.021
δ^*/Δ^*	0.443	0.424	0.409	0.396	0.387	0.379
$\Delta^*(R)/\Delta^*(R-1)$	–	0.574	0.566	0.560	0.554	0.548

Table 1: Optimal ladder parameters against R , in units of σ , at $\text{SNR} = 0$ dB. Obtained by maximizing $C(R, \Delta, \delta)$ directly, so no simulation is involved and the range extends past the swept $R \leq 5$. At $R = 1$ the single pair of thresholds is $\mu/2 \mp \delta$ and Δ is inert.

Across the three panels only the vertical scale changes: normalized by C_∞ the curves coincide to under one percent, and (Δ^*, δ^*) in units of σ barely move. This is Equation (41) showing through – the model enters only through s , which C_∞ absorbs. SNR sets the achievable exponent; R sets what fraction of it is achieved.

Both policies are charged the nominal R , which by Equation (43) overstates what γ^S sends: at $R = 1$ and 0 dB it transmits $\bar{R} \approx 0.59$ bits per sensor, below the one bit γ^V must spend.

6 Appendix

6.1 Full Derivation of the Error Exponent

Error I - $P(\gamma^0(u^{0:N}) = 1|H_2)$

Recalling that the optimal decision rule dictates that $\gamma^0(u^{0:N}) = 1$ if:

$$\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} > Nt \quad (44)$$

Applying the Chernoff bound with $\alpha > 0$:

$$P(\gamma^0(u^{0:N}) = 1|H_2) = P\left(\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} > Nt \middle| H_2\right) \leq E_{H_2} \left[\exp \left(\alpha \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right) \right] e^{-\alpha Nt} \quad (45)$$

From the independence of $y^i|H$:

$$E_{H_2} \left[\exp \left(\alpha \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right) \right] = \prod_{i=1}^N E_{H_2} \left[\exp \left(\alpha \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right) \right] = \prod_{i=1}^N E_{H_2} \left[\left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^\alpha \right] \quad (46)$$

We can calculate a single expectation:

$$E_{H_2} \left[\left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^\alpha \right] = \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i|H_2) \left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^\alpha = \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i|H_2)^{1-\alpha} P^{\gamma^i}(u^i|H_1)^\alpha \quad (47)$$

We now define the policy-dependent Chernoff coefficient $M(\alpha, \gamma^i) \triangleq \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i|H_2)^{1-\alpha} P^{\gamma^i}(u^i|H_1)^\alpha$, the probability of Error I is bound by:

$$P(\gamma^0(u^{0:N}) = 1|H_2) \leq \prod_{i=1}^N M(\alpha, \gamma^i) e^{-\alpha Nt} \quad (48)$$

Error II - $P(\gamma^0(u^{0:N}) = 2|H_1)$

In a similar manner, $\gamma^0(u^{0:N}) = 2$ if:

$$\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} < Nt \iff -\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} > -Nt \quad (49)$$

Applying the Chernoff bound with $\beta > 0$:

$$P(\gamma^0(u^{0:N}) = 2|H_1) = P\left(-\sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} > -Nt \middle| H_1\right) \leq E_{H_1} \left[\exp \left(-\beta \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right) \right] e^{\beta Nt} \quad (50)$$

To align our results, we demand $\beta = 1 - \alpha$, resulting in $\alpha < 1$. From the independence of $y^i|H$:

$$E_{H_1} \left[\exp \left(-\beta \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right) \right] = \prod_{i=1}^N E_{H_1} \left[\left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^{(\alpha-1)} \right] \quad (51)$$

The expectation:

$$E_{H_1} \left[\left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^{(\alpha-1)} \right] = \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i|H_1) \left(\frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \right)^{(\alpha-1)} \quad (52)$$

$$= \sum_{u \in \mathcal{U}} P^{\gamma^i}(u^i|H_2)^{1-\alpha} P^{\gamma^i}(u^i|H_1)^\alpha = M(\alpha, \gamma^i) \quad (53)$$

The resulting bound for Error II is:

$$P(\gamma^0(u^{0:N}) = 2|H_1) \leq \prod_{i=1}^N M(\alpha, \gamma^i) e^{(1-\alpha)Nt}$$

Error probability bound

The probability of error is $P_e = (1-p) \cdot P(\gamma^0(u^{0:N}) = 1|H_2) + p \cdot P(\gamma^0(u^{0:N}) = 2|H_1)$, applying the bounds:

$$P_e \leq (1-p) \cdot \prod_{i=1}^N M(\alpha, \gamma^i) e^{-\alpha Nt} + p \cdot \prod_{i=1}^N M(\alpha, \gamma^i) e^{(1-\alpha)Nt} \quad (54)$$

$$= e^{-\alpha Nt} \left((1-p) + p e^{Nt} \right) \prod_{i=1}^N M(\alpha, \gamma^i) \quad (55)$$

$$= \left(\frac{p}{1-p} \right)^\alpha \cdot 2(1-p) \prod_{i=1}^N M(\alpha, \gamma^i) \quad (56)$$

$$= 2 \cdot p^\alpha \cdot (1-p)^{1-\alpha} \prod_{i=1}^N M(\alpha, \gamma^i) \quad (57)$$

Recalling that $Nt = \log \frac{1-p}{p}$ yields $e^{-\alpha Nt} = \left(\frac{p}{1-p} \right)^\alpha$ and $e^{Nt} = \left(\frac{1-p}{p} \right)$, and $\alpha \in [0, 1]$.

Error Exponent

Recalling the discussion in Section 3.1, we can now bound the error exponent:

$$J_{EE}^N(\gamma^{0:N}) = -\frac{1}{N} \log(P_e) \geq -\frac{1}{N} \log(2) - \frac{1}{N} \log(p^\alpha (1-p)^{1-\alpha}) - \frac{1}{N} \log \left(\prod_{i=1}^N M(\alpha, \gamma^i) \right) \quad (58)$$

Taking the limit will cause the constants to vanish resulting in:

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq \lim_{N \rightarrow \infty} -\frac{1}{N} \log \left(\prod_{i=1}^N M(\alpha, \gamma^i) \right) \quad (59)$$

$$= \lim_{N \rightarrow \infty} -\frac{1}{N} \sum_{i=1}^N \log(M(\alpha, \gamma^i)) \quad (60)$$

Assuming there are K different policies, and N_k sensors using each, meaning $\sum_{k=1}^K N_k = N$. Define $\lim_{N \rightarrow \infty} \frac{N_k}{N} \triangleq c_k$ we get:

$$\lim_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0,1]} \sum_{k=1}^K c_k \log M(\alpha, \gamma^k) \quad (61)$$

6.2 Vanilla Policy: Full Threshold and Chernoff Coefficient Derivation

We first derive the decision threshold for the vanilla setup used in Section 5. By Lemma 1, the optimal decision rule is a threshold τ on the likelihood ratio: the sensor transmits u^2 if

$$L(y) = \frac{f_{y|H_2}(y|H_2)}{f_{y|H_1}(y|H_1)} \geq \tau$$

and u^1 otherwise. Under the Gaussian assumption, $L(y)$ simplifies to

$$L(y) = \frac{f_{y|H_2}(y|H_2)}{f_{y|H_1}(y|H_1)} = \exp\left(\frac{-(y - \mu)^2 + y^2}{2\sigma^2}\right) = \exp\left(\frac{2y\mu - \mu^2}{2\sigma^2}\right)$$

Taking log gives the log-likelihood ratio

$$\begin{aligned} \log(L(y)) &= \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} \geq \log(\tau) \\ \frac{y\mu}{\sigma^2} &\geq \log(\tau) + \frac{\mu^2}{2\sigma^2} \\ y &\geq \frac{\sigma^2}{\mu} \log(\tau) + \frac{\mu}{2} \triangleq \lambda_V \end{aligned}$$

Since the optimal rule corresponds to $\tau = 1$, this reduces to a threshold on y at $\lambda_V = \frac{\mu}{2}$, as used in Section 5.

We now compute the Chernoff coefficient of the vanilla policy:

$$M_V(\alpha, \gamma^V) = P^{\gamma^V}(u^1|H_1)^\alpha \cdot P^{\gamma^V}(u^1|H_2)^{1-\alpha} + P^{\gamma^V}(u^2|H_1)^\alpha \cdot P^{\gamma^V}(u^2|H_2)^{1-\alpha} \quad (62)$$

$$M_V(\alpha) = Q\left(\frac{\mu}{2\sigma}\right)^\alpha \cdot (1 - Q\left(\frac{\mu}{2\sigma}\right))^{1-\alpha} + Q\left(\frac{\mu}{2\sigma}\right)^{1-\alpha} \cdot (1 - Q\left(\frac{\mu}{2\sigma}\right))^\alpha \quad (63)$$

Since $M_V(\alpha) = M_V(1 - \alpha)$, M_V is symmetric about $\alpha = 0.5$; together with its convexity on $[0, 1]$, this means M_V attains its minimum at $\alpha = 0.5$, giving

$$J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0,1]} \log M_V(\alpha) = - \log \left(2 \sqrt{Q\left(\frac{\mu}{2\sigma}\right) \cdot (1 - Q\left(\frac{\mu}{2\sigma}\right))} \right) \quad (64)$$

This is the Chernoff distance bound for the vanilla policy, quoted in Section 5.

6.3 Silent Policy: Full Threshold and Chernoff Coefficient Derivation

Recall the silent setup of Section 5. We introduce two thresholds $\tau_1 < \tau_2$ on the likelihood ratio: the sensor sends u^1 if $L(y) < \tau_1$, u^2 if $L(y) > \tau_2$, and stays silent (u^S) otherwise:

$$R_1 = \{y \in \mathcal{Y} : L(y) < \tau_1\}, \quad R_S = \{y \in \mathcal{Y} : \tau_1 \leq L(y) \leq \tau_2\}, \quad R_2 = \{y \in \mathcal{Y} : L(y) > \tau_2\}$$

Since $L(y)$ is increasing in y (Appendix 6.2), these map to thresholds on y : with $\lambda_i \triangleq \frac{\sigma^2}{\mu} \log(\tau_i) + \frac{\mu}{2}$ for $i = 1, 2$ and $\lambda_1 < \lambda_2$,

$$R_1 = \{y < \lambda_1\}, \quad R_S = \{\lambda_1 \leq y \leq \lambda_2\}, \quad R_2 = \{y > \lambda_2\}$$

giving the decision rule γ^S :

- $P^{\gamma^S}(u^1|H_j) = \int_{R_1} f_{y|H_j}(y|H_j)dy$
- $P^{\gamma^S}(u^S|H_j) = \int_{R_S} f_{y|H_j}(y|H_j)dy$
- $P^{\gamma^S}(u^2|H_j) = \int_{R_2} f_{y|H_j}(y|H_j)dy$

Evaluating these integrals under the Gaussian model gives:

- $P(u^1|H_1) = P(y < \lambda_1|H_1) = 1 - Q(\frac{\lambda_1}{\sigma})$
- $P(u^S|H_1) = P(\lambda_1 \leq y \leq \lambda_2|H_1) = Q(\frac{\lambda_1}{\sigma}) - Q(\frac{\lambda_2}{\sigma})$
- $P(u^2|H_1) = P(y > \lambda_2|H_1) = Q(\frac{\lambda_2}{\sigma})$
- $P(u^1|H_2) = P(y < \lambda_1|H_2) = 1 - Q(\frac{\lambda_1 - \mu}{\sigma})$
- $P(u^S|H_2) = P(\lambda_1 \leq y \leq \lambda_2|H_2) = Q(\frac{\lambda_1 - \mu}{\sigma}) - Q(\frac{\lambda_2 - \mu}{\sigma})$
- $P(u^2|H_2) = P(y > \lambda_2|H_2) = Q(\frac{\lambda_2 - \mu}{\sigma})$

λ_1 and λ_2 are in general independent, but since the two Gaussians share variance σ^2 and differ only in mean, the optimal boundaries are symmetric about the midpoint $\frac{\mu}{2}$. We therefore parametrize them by a single variable $\delta > 0$:

$$\lambda_1 = \frac{\mu}{2} - \delta, \quad \lambda_2 = \frac{\mu}{2} + \delta$$

reducing the optimization to a single free parameter. Denoting

$$q_+ \triangleq Q\left(\frac{\mu/2 + \delta}{\sigma}\right), \quad q_- \triangleq Q\left(\frac{\mu/2 - \delta}{\sigma}\right)$$

and substituting λ_1, λ_2 into the probabilities above gives:

- $P(u^1|H_1) = 1 - q_-, \quad P(u^S|H_1) = q_- - q_+, \quad P(u^2|H_1) = q_+$
- $P(u^1|H_2) = q_+, \quad P(u^S|H_2) = q_- - q_+, \quad P(u^2|H_2) = 1 - q_-$

as used in Section 5. We now compute the Chernoff coefficient of the silent policy:

$$M_S(\alpha, \gamma^S) = P^{\gamma^S}(u^1|H_1)^\alpha P^{\gamma^S}(u^1|H_2)^{1-\alpha} + P^{\gamma^S}(u^S|H_1)^\alpha P^{\gamma^S}(u^S|H_2)^{1-\alpha} \quad (65)$$

$$+ P^{\gamma^S}(u^2|H_1)^\alpha P^{\gamma^S}(u^2|H_2)^{1-\alpha} \quad (66)$$

$$M_S(\alpha) = (1 - q_-)^\alpha q_+^{1-\alpha} + (q_- - q_+)^\alpha (q_- - q_+)^{1-\alpha} + q_+^\alpha (1 - q_-)^{1-\alpha} \quad (67)$$

Since $P(u^S|H_1) = P(u^S|H_2) = q_- - q_+$, the middle term collapses to $q_- - q_+$ regardless of α :

$$M_S(\alpha) = (q_- - q_+) + (1 - q_-)^\alpha q_+^{1-\alpha} + q_+^\alpha (1 - q_-)^{1-\alpha} \quad (68)$$

Swapping $\alpha \leftrightarrow 1 - \alpha$ leaves the constant term unchanged and exchanges the remaining two, so $M_S(\alpha) = M_S(1 - \alpha)$; together with convexity on $[0, 1]$, this means M_S attains its minimum at $\alpha = 0.5$, as in the vanilla case, giving

$$J_{EE}^N(\gamma^{0:N}) \geq - \min_{\alpha \in [0, 1]} \log M_S(\alpha) = - \log \left((q_- - q_+) + 2\sqrt{(1 - q_-)q_+} \right) \quad (69)$$

This is the Chernoff distance bound for the silent policy, quoted in Section 5.

6.4 Optimal Silence Threshold

The objective. Substituting $q_{\pm} = Q(\frac{\mu/2 \pm \delta}{\sigma})$ into Equation (??):

$$M_S(\delta) = \left[Q\left(\frac{\mu/2 - \delta}{\sigma}\right) - Q\left(\frac{\mu/2 + \delta}{\sigma}\right) \right] + 2\sqrt{\left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right] Q\left(\frac{\mu/2 + \delta}{\sigma}\right)}. \quad (70)$$

Using $1 - Q(x) = Q(-x)$, equivalently

$$M_S(\delta) = 1 - Q\left(\frac{\delta - \mu/2}{\sigma}\right) - Q\left(\frac{\delta + \mu/2}{\sigma}\right) + 2\sqrt{Q\left(\frac{\delta - \mu/2}{\sigma}\right) Q\left(\frac{\delta + \mu/2}{\sigma}\right)}, \quad (71)$$

giving $M_S(0) = 2\sqrt{q(1-q)} = M_V$ and $M_S(\delta) \rightarrow 1$ as $\delta \rightarrow \infty$.

Differentiation. $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$, so $Q'(x) = -\phi(x)$ with $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. The inner derivatives are $\mp 1/\sigma$, so

$$\frac{d}{d\delta} \left[Q\left(\frac{\mu/2 - \delta}{\sigma}\right) - Q\left(\frac{\mu/2 + \delta}{\sigma}\right) \right] = \frac{1}{\sigma} \left[\phi\left(\frac{\mu/2 - \delta}{\sigma}\right) + \phi\left(\frac{\mu/2 + \delta}{\sigma}\right) \right]. \quad (72)$$

For $f(\delta) \triangleq \left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right] Q\left(\frac{\mu/2 + \delta}{\sigma}\right)$,

$$f'(\delta) = -\frac{1}{\sigma} \phi\left(\frac{\mu/2 - \delta}{\sigma}\right) Q\left(\frac{\mu/2 + \delta}{\sigma}\right) - \frac{1}{\sigma} \phi\left(\frac{\mu/2 + \delta}{\sigma}\right) \left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right], \quad (73)$$

and $\frac{d}{d\delta} 2\sqrt{f} = f'/\sqrt{f}$. Combining:

$$\sigma \frac{dM_S}{d\delta} = \phi\left(\frac{\mu/2 - \delta}{\sigma}\right) + \phi\left(\frac{\mu/2 + \delta}{\sigma}\right) - \frac{\phi\left(\frac{\mu/2 - \delta}{\sigma}\right) Q\left(\frac{\mu/2 + \delta}{\sigma}\right) + \phi\left(\frac{\mu/2 + \delta}{\sigma}\right) \left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right]}{\sqrt{\left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right] Q\left(\frac{\mu/2 + \delta}{\sigma}\right)}}, \quad (74)$$

which is Equation (??) written out.

Factorization. Multiplying by $AB > 0$:

$$\begin{aligned} (\phi_- + \phi_+)AB - (\phi_-A^2 + \phi_+B^2) &= \phi_-A(B - A) + \phi_+B(A - B) \\ &= (B - A)(\phi_-A - \phi_+B). \end{aligned} \quad (75)$$

Since $B - A > 0$, setting $M'_S = 0$ forces $\phi_-A = \phi_+B$. The density ratio in closed form,

$$\frac{\phi_-}{\phi_+} = \exp\left(\frac{(\mu/2 + \delta)^2 - (\mu/2 - \delta)^2}{2\sigma^2}\right) = e^{\mu\delta/\sigma^2}, \quad (76)$$

which gives Equation (??).

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